

From Pixels to Presence: Telepresence and the Construction of Meaningful Spaces in Gaming Cultures

Process Book

Travis Olavi Niskanen Ladore

Game Design / 40944384



Preface

This is my second attempt at finishing my studies at UE. My first attempt was unsuccessful due to many unfortunate things happening over that semester that made it quite difficult to focus on getting a thesis paper and project done. However, after taking a semester break and coming back to attempt and get my studies done, I came with a new found energy that I never had before. I had to come up with a new topic since its a new attempt and my topic before is no longer valid. I asked my second supervisor, Wolf, for some advice on what I was thinking, and he mentioned a term called telepresence and it peeked my interest. After delving headfirst into the subject I thought it was a pretty cool subject to work on and wanted to make my thesis about it.

My topic is on Telepresence, the feeling of being present somewhere you are not currently physically inhabiting. The reason why I wanted to study this topic is because of my friends back home in the United States. These friends are very close to my heart and one way we keep in contact and spend time together is to inhabit virtual spaces together where we can let our personalities shine. I thought this was very important to research because of how close I feel with my friends and because of how we form memories in virtual spaces that seem tangible and mean something to us. So to try and justify that I wanted to see what was going on and dedicate my studies to it.

Beyond that, video games have been a medium of entertainment that has been in my life for as long as I can remember and I have an entire vault of memories in my head that take place in virtual spaces. Looking back on gaming experiences and not necessarily remembering my physical surroundings but instead remembering what was around my virtual avatar in a game and what I was experiencing in the game world.

With the world becoming more digitized by the day and with more of us exploring virtual spaces, I wanted to attribute meaning to that time spent so I can better understand myself and the

people who enjoy this. With that, lets go into the thought process behind making the game and the other things in life that I needed to handle to get this project done and presentable.

This PDF is a collection of my personal handwritten notes on this as well as images of things during my project period.

The Main Game idea

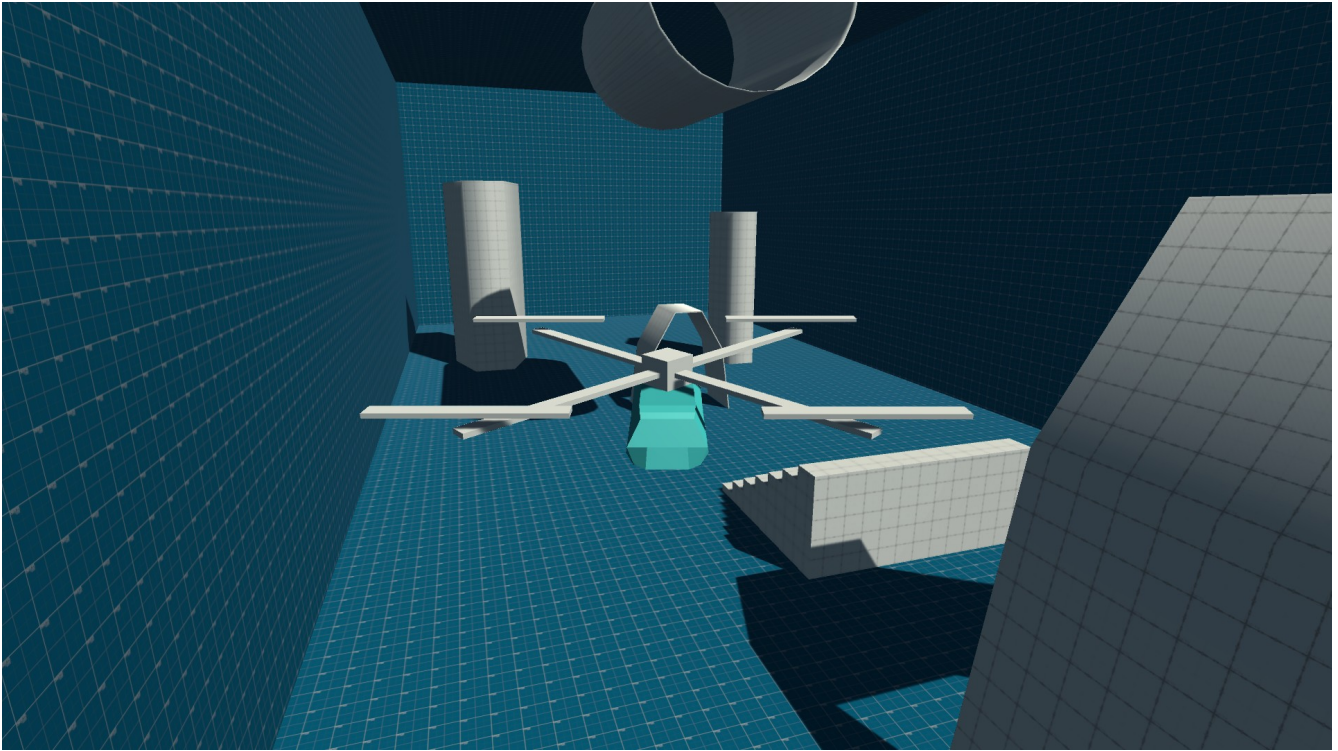
I wanted the game to be a simple space that lets the player feel a sense of awe and feel present. I also wanted to keep it simple when it comes to the mechanics and decided for a simple pick up mechanic and an interact mechanic for the first person player character. I also wanted to make It so multiple people can play at once and experience the virtual space together. If that proved to be too difficult then I would make it so a player can leave some sort of impact on the world and then the next player can come along and find little remnants of other people being there. I wanted the game to be a 200x200 meter space with areas that highlighted different ideas related to telepresence. I wanted the game to have a distinct visual style and I wanted to be able to make most of the game assets myself.

How it started

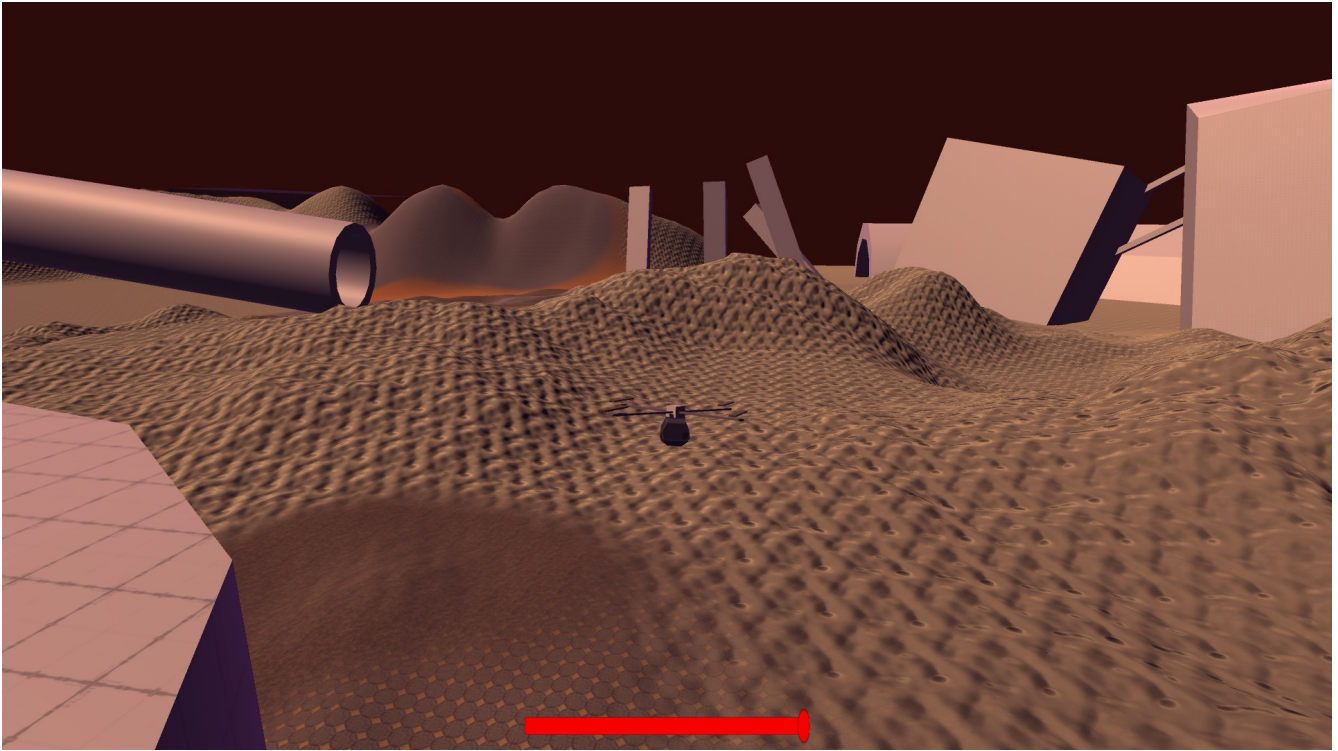
Tiny-Drone

I started working on different projects throughout my time away from University because I wanted to keep my unity skills sharp and I also just wanted to experiment with different things that felt good.

One of the first projects I worked on during my semester off was a game called Tiny-Drone. Tiny Drone was a simple project about flying a drone around an apocalyptic landscape, piloted by someone who can only inhabit a bunker deep underground because of the toxic outside environment.



This is the first iteration of the drone. I found a script online that handled all of the drone mechanics and I was able to edit variables how I saw fit. I made a simple course to fly through and this can be played on my itch page, <https://slsuu.itch.io/> and the repository link is <https://github.com/Travis-Ladore/TinyDrone>. After playing around with this and having quite a bit of fun just flying around and trying to not hit any obstacles, I decided to try and make the environment it would fly in and came up with the image below. Its simple and its fun and I had the feeling of being present while flying the drone and I really wanted to explore more of an open space and see how it was flying around bigger obstacles and to see a large sprawling landscape.



I also added a bunch of ambient sound to the project and playing it feels like a tangible experience. I also really liked how it made me feel a sense of dread. I stopped working on this and kept it as is, but always kept it in mind for the project because early studies of telepresence talked about telerobotics and I think controlling a player character that is controlling a drone was sort of a meta thing to do. There was different layers of control in this this game and if I used similar mechanics for the thesis project it would, hopefully, allow the player to feel a bit more present in the virtual world and experience the space in more than one way.

Tiny Drone was not a game I worked on during my thesis period but its important to understand for the final version of the game that is being submitted.

MatteBlack

The next small project I worked on that was actually during my thesis period where I was experimenting with different things to pilot around a virtual world. This project was called Matte Black https://github.com/Travis-Ladore/Matte_Black Matte Black wasn't really an idea to make anything, it was simply to test out a mechanic I really wanted to implement into the final game as well.

The mechanic I wanted to implement was a spotify to unity integration where I could pull up my spotify through a unity game and then be able to play and control the music through the game. The reason for this is because I wanted players in my thesis project to be able to listen to music that they choose and hopefully connect to the space a bit more.

I got the spotify to unity integration working but with some bugs that made it more inconvenient than anything. It works, but when playing on PC I needed my spotify client open and I couldn't actually play the spotify music through unity, it simply controlled my spotify session. I tried to look up more on the issue however, I was also working on the written part of my thesis at this time and just left the project on hold.

Starting the Thesis Project

I started the project by making a repository on my GitHub account called, "Metal". The only reason for this is because I was listening to metal music at the time. <https://github.com/Travis-Ladore/Metal> I made this project after Matte Black and for this project, I wanted to try and rediscover my gaming history. I started playing games at a young age and one of the first memories I have with gaming is a game called Final Fantasy VIII. I remember when I knocked on my brothers door to ask to watch him play PlayStation 1 and I remember walking in and seeing the intro sequence to Final

Fantasy VIII and being stunned. It was one of the coolest things I had ever seen at the time and I would suggest you watch it here, https://www.youtube.com/watch?v=8CAVrwaZI80&ab_channel=FINALFANTASY.

I also remember playing so many other games on the PlayStation 1 with my brother and the visual style is always something that's going to get an emotional response from me so I really wanted to use this visual style for my project. Here are some examples



Tekken 3



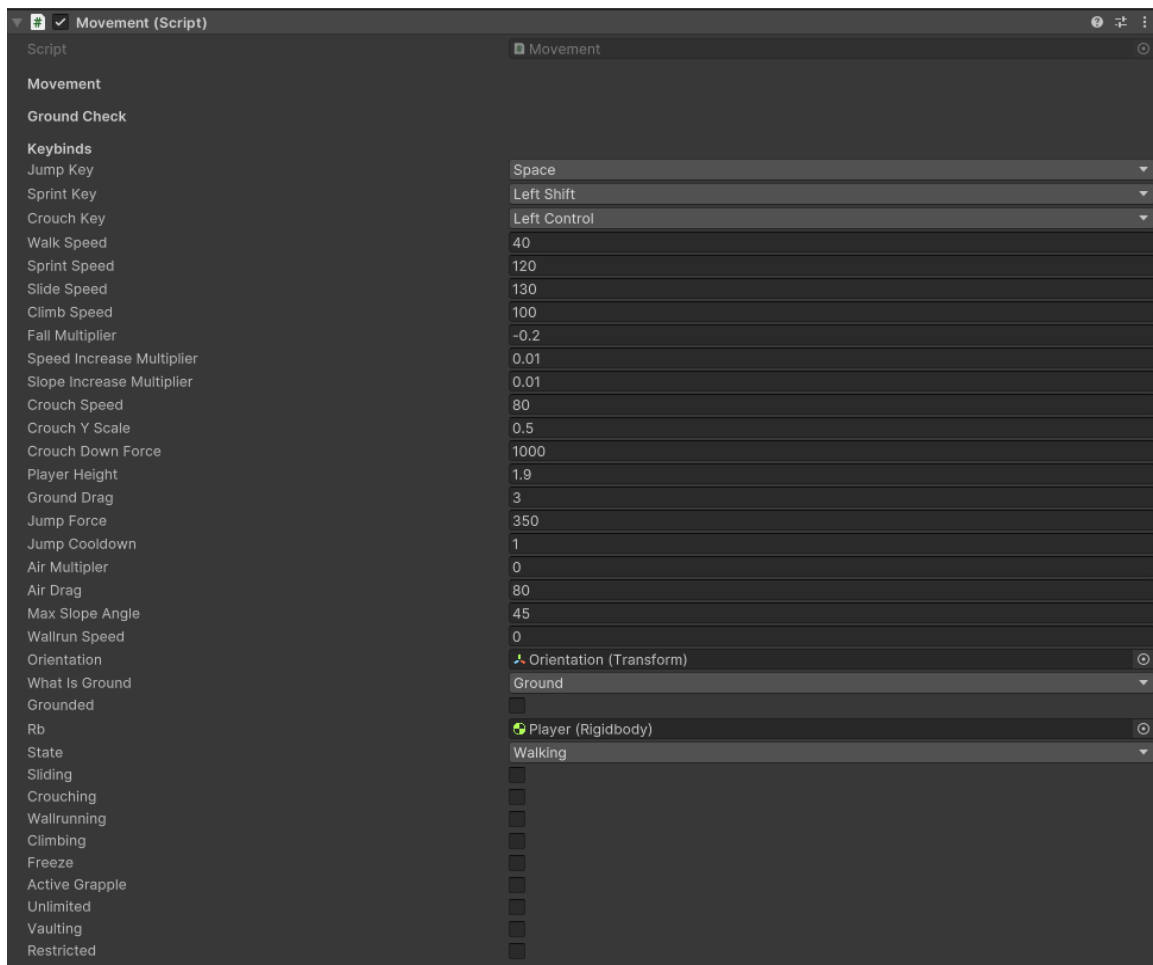
GoldenEye



Final Fantasy VIII

These games for the time were pushing the boundaries of 3D graphics and after replaying them to get inspiration, the visual style still has its charm and gives a certain feeling of nostalgia. I also wrote in my thesis that I wanted to give the game a unique visual style and this is the one I chose.

It was easy to find a unity package that allowed me to get this visual style without too much work on writing shaders. Because I have absolutely no experience with that and didn't want to take that deep dive for this project. I decided to make the project with the 2022.3.13fl version since its long time support and I did not have that much room on my hard drives so I just used what I had. I made the repository and I quickly added a first person movement script that I have been using for a while now that I wrote after watching this tutorial series https://www.youtube.com/watch?v=f473C43s8nE&ab_channel=Dave%2FGameDevelopment

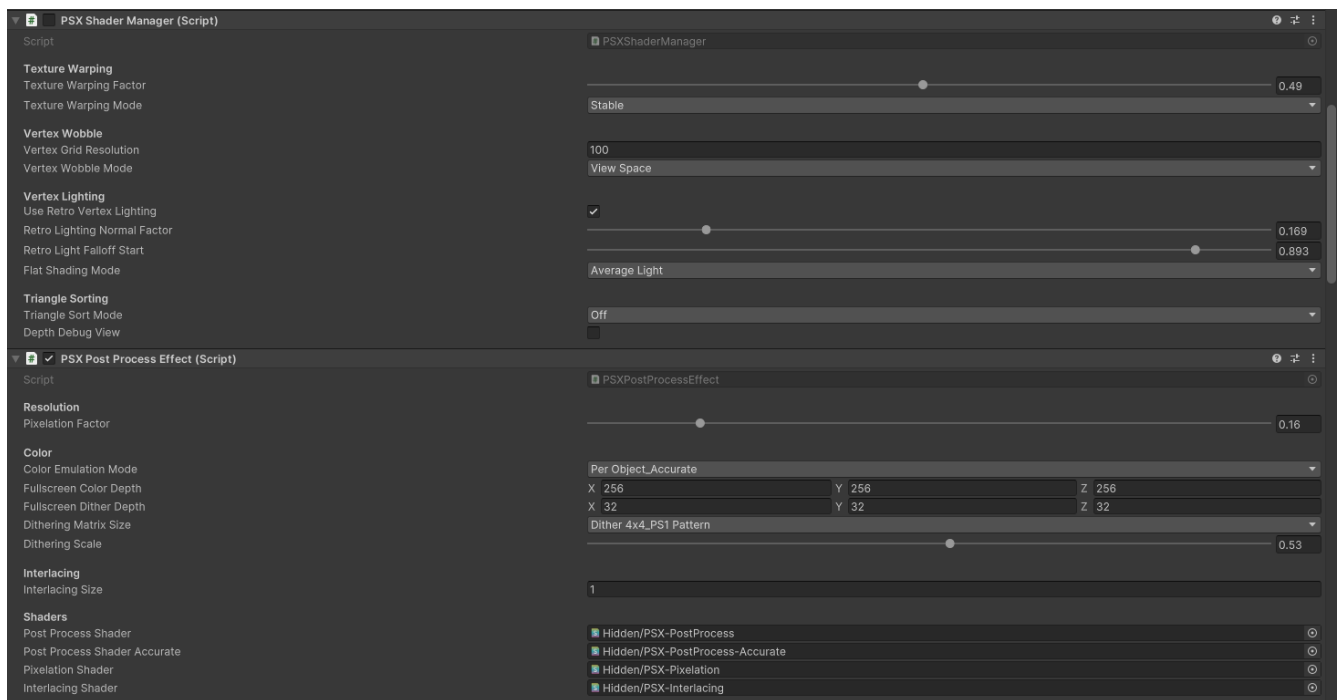


This First person controller uses rigid-body physics and I always liked using rigid-body physics for the player so I could just use unities built in physics calculations. This controller also gives quite a lot of flexibility for me since I can change everything.

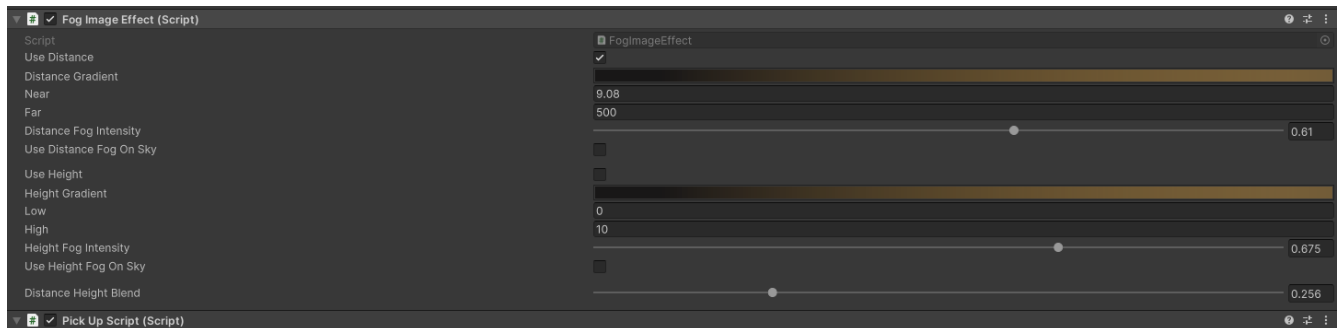
Since this movement script has the freedom to edit, I can use it for a multitude of projects without needing to code all of it all over again. Its nice to think about how Csongor used to tell us to be smart about how we should make flexible assets that can be used for different things. So thanks for that!

So then I combined the PSX shader unity Package with the first person controller to make a simple environment.





PSX Shader in Unity

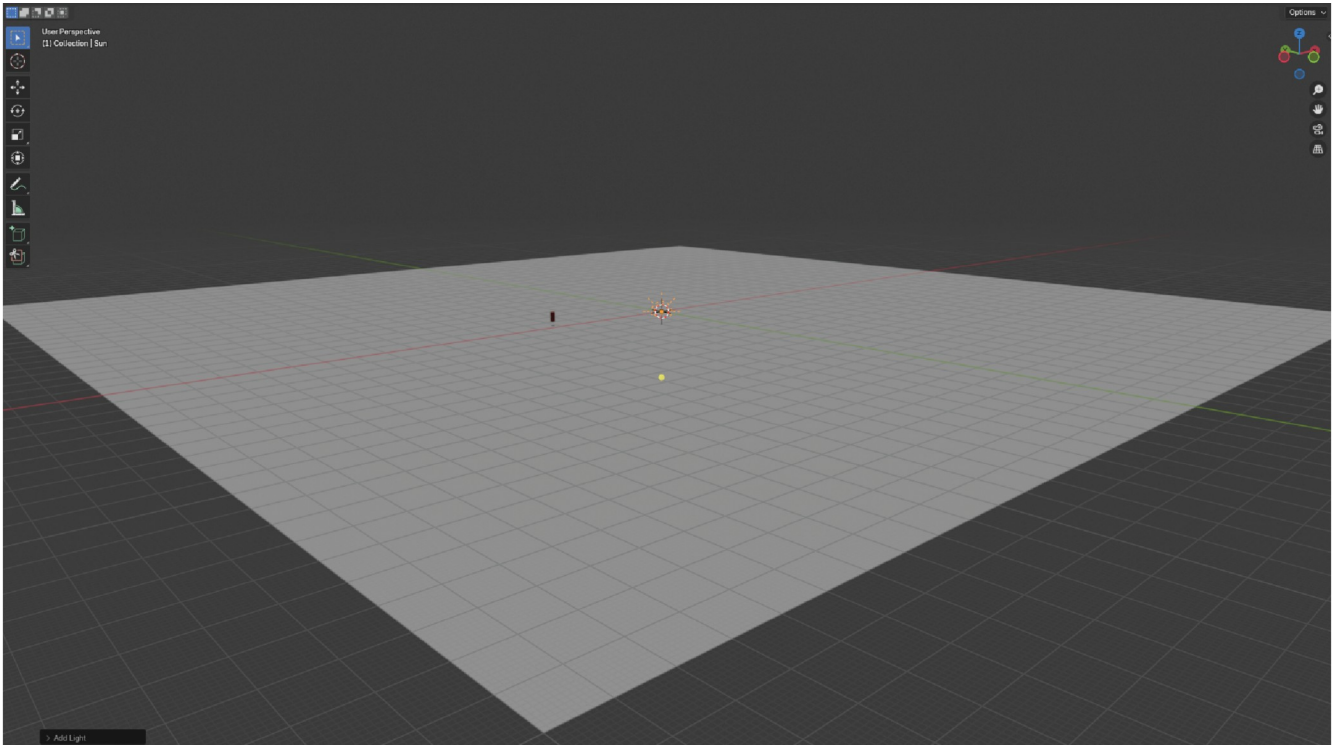


PSX Shader in Unity

The PSX shader pixalates whatever the camera sees and also adds a dithering effect as well as a fog effect. Everything was changeable as well which is super nice and I experimented with a lot of different intensities, fog colors, and other settings. It took me a while to find out what I liked and already I thought everything looked unique and also allows me to make assets low poly and textures smaller. All in all, this was quite a good start towards the game. After this, it was time to make assets.

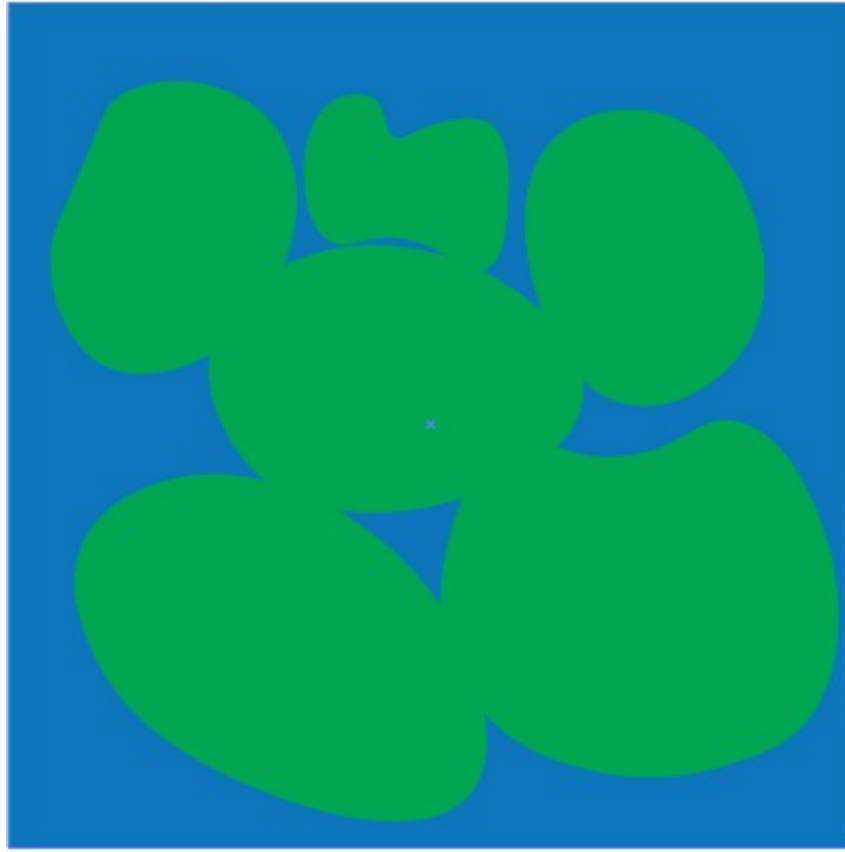
Making Assets

I always wanted to make something with a unique visual style with most of the assets made by me. My 3D experience is still quite low and I wanted to challenge myself to practice my 3D work and create this space. I started with thinking about the terrain.



200x200 meter plane in blender

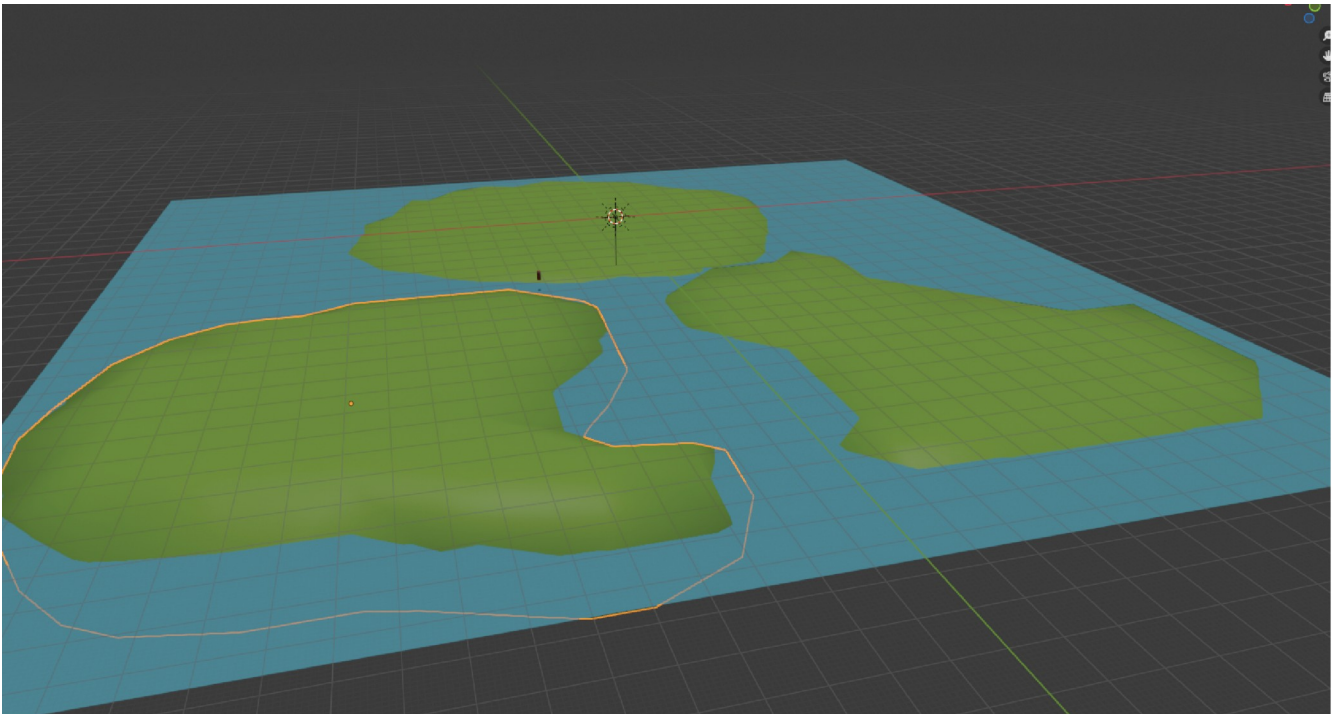
I originally thought about having a 200 meter x 200 meter playing area for the game to keep things small but also allow players to explore a little bit. Around this time, Unity 6.0 came out and I was interested in testing things out with the new engine. Before that though, I drafted different layouts for my game



Here's one of those original ideas. I had the 200x200 plane act as a water barrier and the different green spots act as islands. This was also after spending some time playing Webfishing and I was really inspired by the overall design of the game.



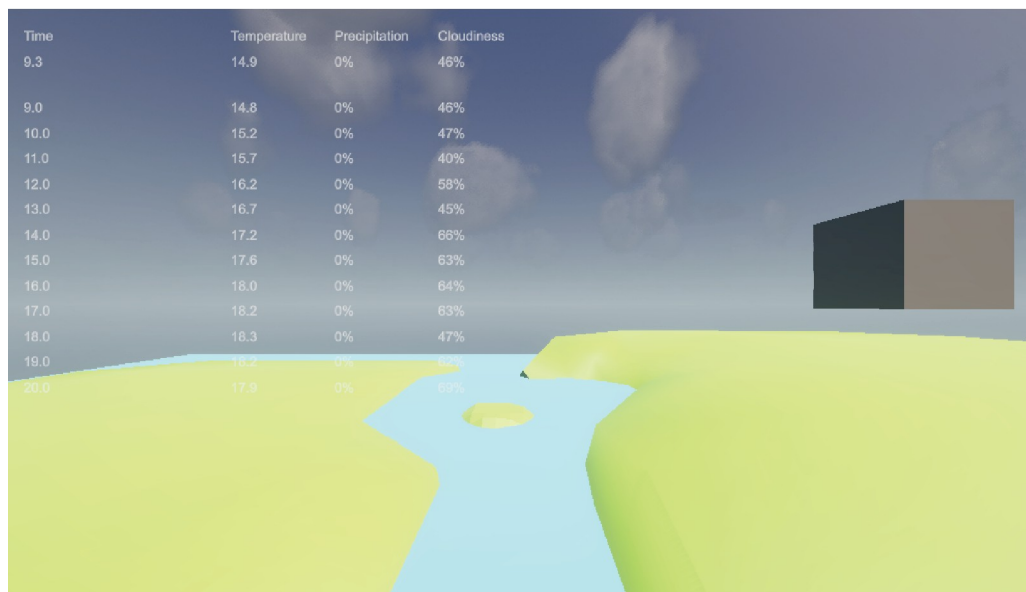
The game has a very distinct style and originally it was something I really wanted to emulate. I went as far as creating a new project in Unity 6 and made a water shader for the plane.



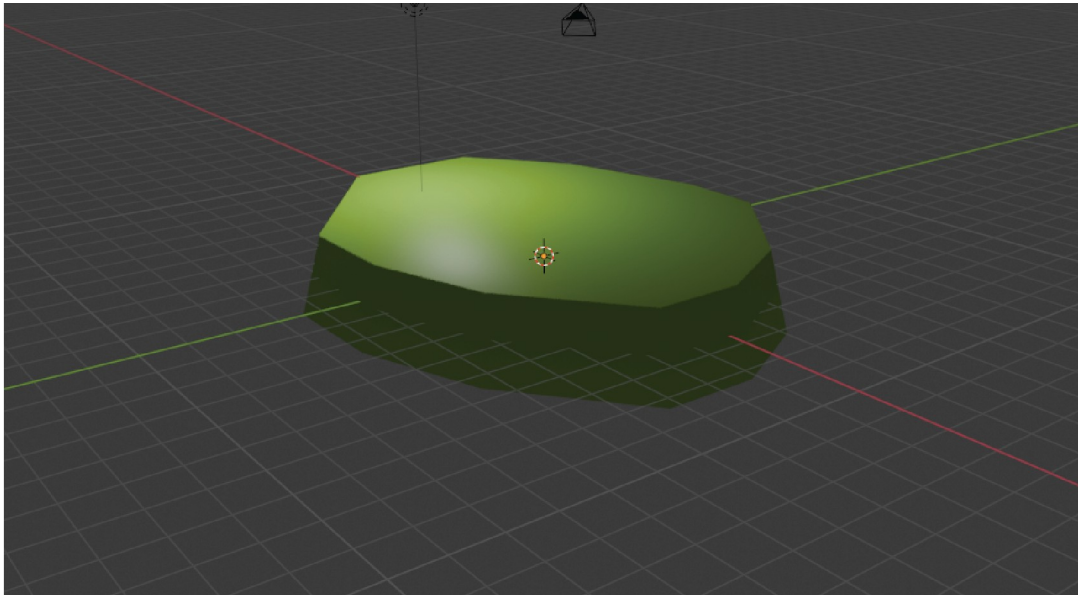
An early concept for the level

I liked how this early example looked in blender and moved it to unity

sds



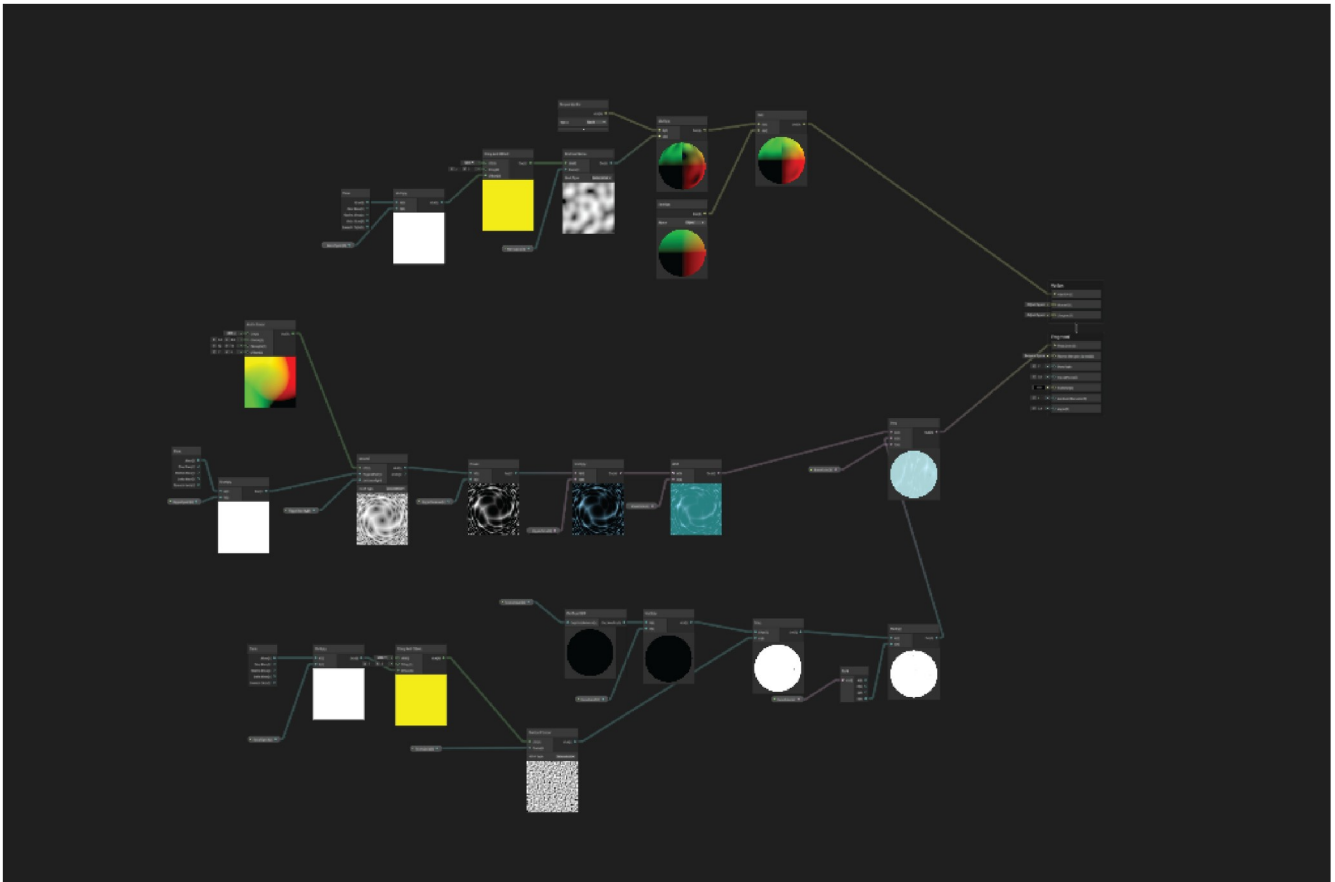
I also used another package I had which makes volumetric clouds making this. When first looking at it, there's a lot of things I don't like. I don't like the low poly style with the HD clouds, I don't like the soft angles on the terrain going into the water and I'm not using the PSX shader that I wanted to use since I could not get it to work with Unity URP. Though, I took one last attempt to see if working with URP in unity 6 was something I wanted to keep doing.



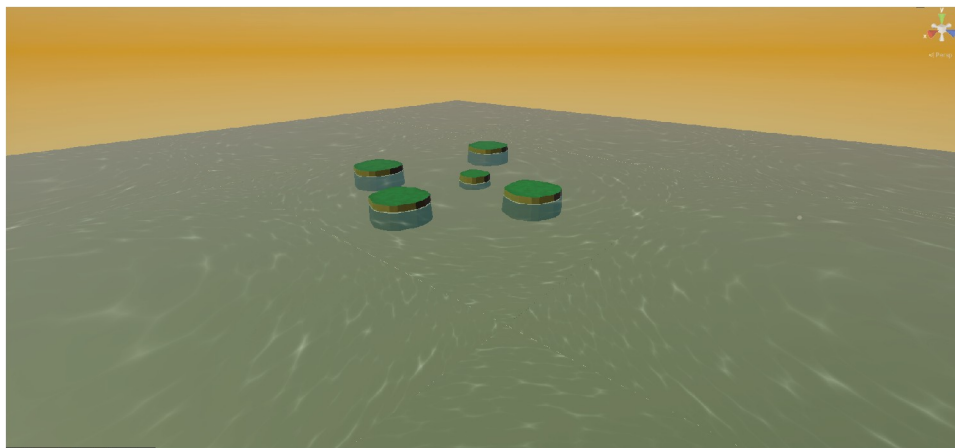
I made a new land object that was lower poly and was also closer to what I wanted.



I then made a simple texture for it and put it in unity to see how the islands would look. I combined these little islands with a water shader that I made by following a tutorial on it.



Water Shader graph in Unity 6



It ended up looking like this and it brought up a lot of questions for me. What would happen if the player fell in the water? I could just put colliders around the islands and make sure that players couldn't

fall off but that felt cheap. You're surrounded by water and can't swim? Not really good game design. I also didn't have a real reason to have islands surrounded by water. I really just wanted to have these islands act as a natural barrier to different areas which have different focuses related to telepresence. So I scrapped the idea entirely and went back to my original idea.

An unexpected Hurdle

At the end of September to the beginning of December I had some health issues and had to be in the hospital and was unable to work. Around this time, I was having trouble breathing and even mild exertion would make me have to catch my breath. It was a hard time but I still had my project in my mind. It was hard not to think about the time I was losing but I did what I could when I could. Even if that was playing games with the same art style while I rested. I thought about a lot of the games I played that led me to want to make a game on this subject. I thought about the server that my best friend and I have in a game called Valheim and how that's spanned over 4 years now. I thought about my early experiences in gaming and I thought about the games that I play with my current friend group. I thought it was interesting how much time I spent inhabited virtual worlds and I took a sort of detox from any type of screen time and just focused on getting better in the hospital.

I had a lot of time to reflect on my life and when my health issues got the best of me it was difficult to find motivation on the project. That being said, one thing I did consistently was watch YouTube. One youtuber that I watched a lot was Jacob Geller and if you have not seen him, I would highly recommend him if you're interested in long video essays about media, specifically gaming. One video that spoke to me was his video titled "A thousand ways of seeing a forest"

https://www.youtube.com/watch?v=JgQ_buy8xJ0&ab_channel=JacobGeller In the video he talks a lot about forests of games are an extension of the people who created them. It was poetic and it gave me a lot of inspiration towards my project.

There was a day where I had gotten back from the hospital after a few weeks and came to my computer and opened my thesis paper to read the title “Telepresence and the Construction of Meaningful Spaces in Gaming Cultures” and thought about how to attribute personal meaning to this and how I could create my own virtual forest. I wanted to incorporate different things and different themes of my actual life into the visual design of things and I was artistically inspired. I started by working on trees which was simple enough. I used a program called Tree It, which is available on steam, to make the foliage in my game.



Tree It Is super easy to work with and it comes with some tree and leaf textures already that I ended up using for the duration of the project. The leaves moved and it was so easy to make unique looking trees. Putting them into the game was cool and all but I still wasn't completely sure on the terrain of the game. I thought that the best thing would be to just use a 200x200 meter terrain and paint that terrain

with textures to make it look nice. I would also just use mountains as natural barriers so the player doesn't go into unintended places. I found some ground textures on opengameart and went ahead and added them to the project to make some terrain.

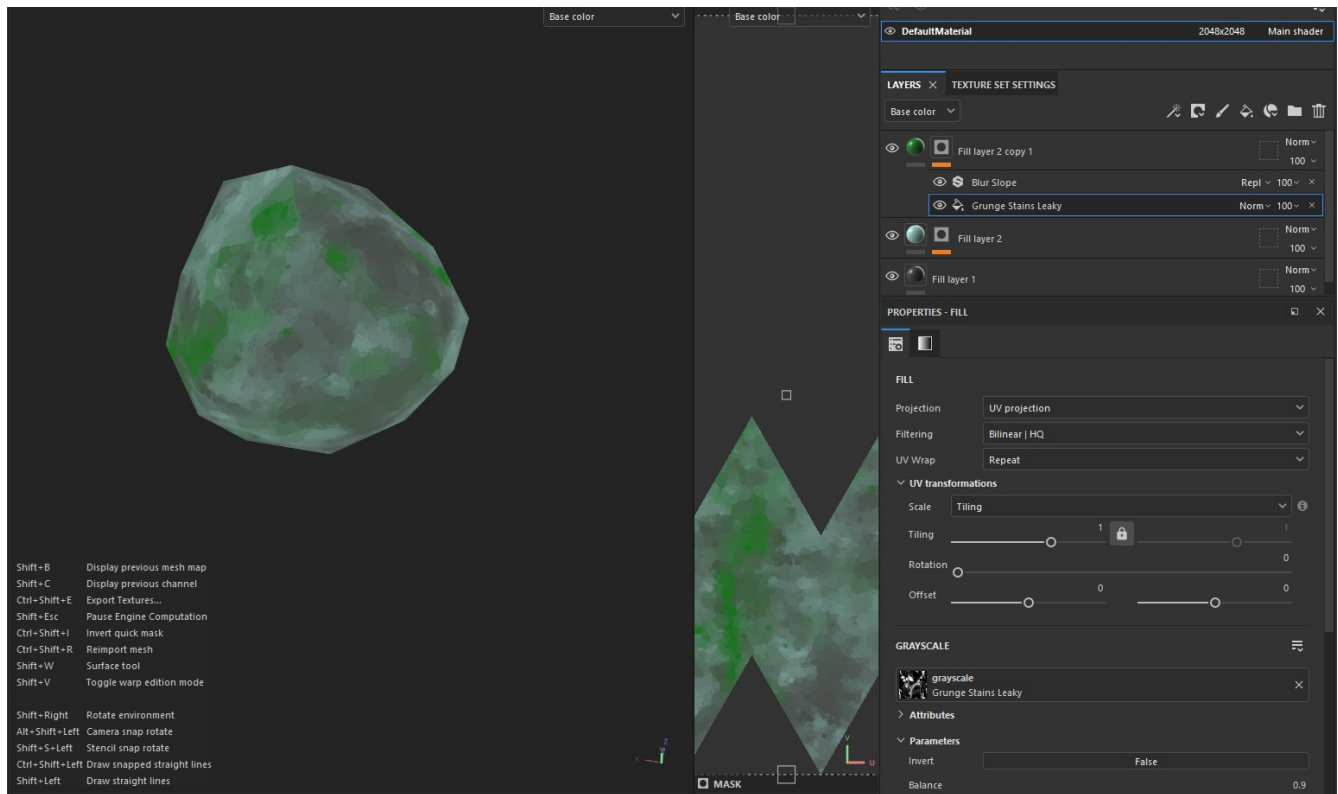


Here's the terrain with some other assets thrown in to make it look nice. This combined with the PSX shader made for a really unique look.



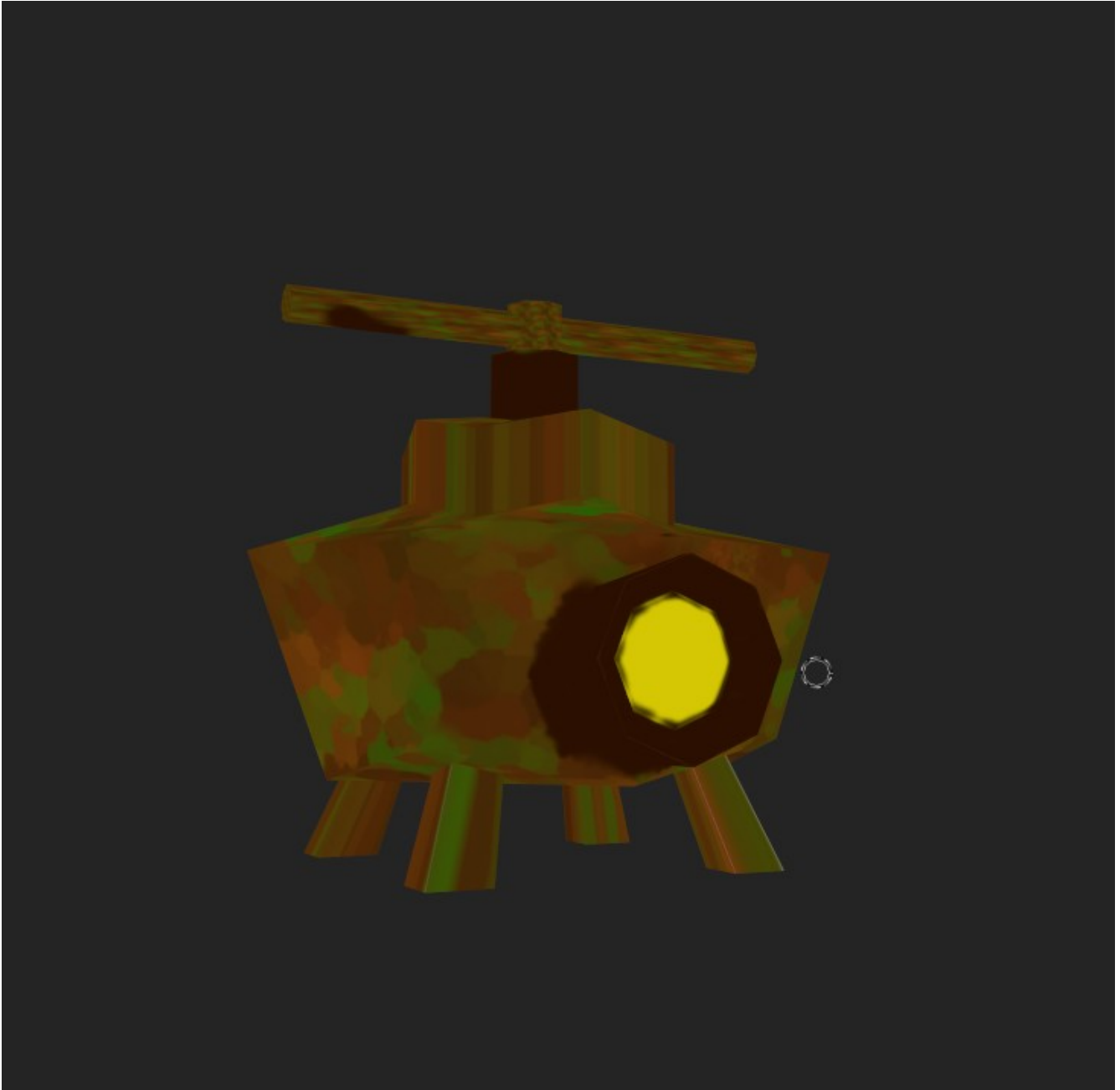
Working in 3D

Before this, I didn't have a lot of 3D modeling experience but since I really wanted to get better at that, I made sure to create as much as I could for this game. I ended up with a hard drive full of different FBX files and blender projects. For me, the hardest part was actually texturing things because I still hadn't really understood how to make good UV's where textures wouldn't warp. I came across a tutorial for adobe substance painter that was easy to follow and was about making a style similar to a studio ghibli film.



The process involved making a few fill layers with black masks and adding a texture to that black mask which would then leave a pattern with the color you choose. It was really easy to make stylized textures with this method and it also helped me make a lot of assets really fast. I would have an idea, make a really quick low poly model in blender and then texture it to fit this style. Then put it into my Main

Unity scene to build a world. Overall it was really fun to make a bunch of models for this and to get more comfortable with the process.



It was about 3 weeks out to the due date and I realized I didn't really have a lot in terms of actual game design and had a lot of assets but nothing to do with them as the player. This was also about the time

where I was spending a lot more time at work and trying to balance work and school was quite difficult.

Balancing life and game design

Once my health was in order and I was stable and I was able to find my strength, I put all that into working on the project while working 30-40 hours at my kitchen job. I would say the hardest part of this entire project was balancing work and school. After just recovering from my health issues, I had to make up for a lot of lost hours at work and needed to pay bills. My restaurant is also busy ALL of the time and its never a day without at least making a hundred dishes for morning brunch. Sometimes after an eight hour shift on the line with only a 30 min break, I went home and just tried to recover from the day, getting very little done. However, after spending time in the kitchen and getting used to how things flowed I was making dishes without thinking about it and I spent a lot of time lost in the idea of my project during rush hour when we were drowning in orders.

It was like that for a while until I found my full stride again and was even able to start regularly going to the gym again. After that I went right back to work on the project. I went back to the drawing board in terms of the mechanics. I didn't have a lot of time left but I still wanted my game to reach the intended effect. I made the hard decision to cut the multiplayer aspect because I didn't want to try and make too many multiplayer features because it would take a lot more time than a single player experience. I did however see a way around this by thinking about a save function where things that players did before would stay in the world and during the exhibition, the world would have more players leave their trace behind and hopefully helping players feel present in a world.

I think I really wanted to feel as if the space is not only serene but not unreachable and that other people were there before you. I stuck with the pick up mechanic as well as the interact mechanic because those in themselves can have a lot of other things added to it. For example, I added a pressure plate mechanic so the player can pick something up and activate the plate. The pressure plate can be

used for a multitude of reasons and so can the pickup mechanic. For the interact mechanic, I wanted to make a stone that players would walk up to and be able to write something on. When interacted with, I wanted the player to be met with a text field, prompting something and giving the player the freedom to write whatever. That way, the player that plays next can come behind them and see the impact anonymously.

Dealing with crunch

About two weeks out from the project due date, I spent a lot of time just making 3D assets to try and fit a visual style. One that I think looks really nice but I still didn't really have many mechanics to implement and was quickly running out of time. I also was spending more and more time at work trying to make enough money to pay rent. I had to scrap a few ideas and even getting towards the final due date, I was not very satisfied by what I had and had to make due to still get the intended feeling of being present in a virtual world. Well I thought to myself that I could lean into the visual style and make that the whole experience. At the bare minimum, I wanted players to inhabit a space that means something to me and in turn, make the player feel something special. So I got to work with even more models and really tried to make something special in the time left.

I had to rely on past projects and code that I already have to make some type of game play and what I ended up with is not too bad. I had a hard time with crunch because of the time I lost while in the hospital but I also didn't ask for an extension for reasons unknown. In the end it was a lot of stress and I still would like to make a different build for the exhibition that people can also play but I'm not so sure that's allowed. Dealing with crunch is always going to be difficult but it seems like when I look at the repository commits, I get the most done when I'm under pressure.

Conclusion and end of project thoughts

Towards the end of the project I was a bit burnt out. I had worked so much on making the visual style stand out that I was left with a project that had little to no mechanics. This was unfortunate because the mechanics I did want to implement were quite cool in concept. I think this was my hardest semester because it was all about me and my ability to get things done. I had a good head start on the project but losing a lot of time towards health issues was the hardest thing. At some point, in the hospital, I doubted whether or not I would get anything done but in the end I have something that is an extension of myself and that gives the virtual space enough character to feel present in. At least, that's what my friends who played told me and that was a really nice thing to hear. I was happy with my early decision on the size of the map because that allowed to stay focused on a small area and relieved some of the stress.

One thing that I learned was how to properly work on a game as to not get too burnt out and how to balance life and work. I even took a lot of inspiration from work and daily life to incorporate things into the game. Overall, this project was a wonderful experience and I will most likely continue to work with the same repository because I have become attached to the visual style.

One mechanic that I was happy to implement was the drone mechanic from Tiny Drone. If anything I think this was the most important mechanic to include because of the origins of Telepresence being with telerobotics and it was also just fun to move around the scene with the drone.

In the end, the game turned out more as a meditative experience where players can just wander around and enjoy the virtual space with little interaction. For me, this is okay, when people played, they said they were able to see the idea and also feel a certain kind of presence when playing the game. What I learned is that you really don't need a lot to help a player feel present in your virtual world, an

amazing start is to create an environment that is captivating and build from there. I hope to continue to work on this project idea in the future but for now it's time to turn it in and leave it be. Thanks for reading.

Repository link: <https://github.com/Travis-Ladore/Metal/tree/main>

Itch.io Link: <https://s1suu.itch.io/telepresence-thesis-project>

-Travis Ladore